

## ARTIFICIAL INTELLIGENCE

COURSE TITLE: **COMPUTER VISION**

COURSE CODE: **AIT 322**

### COMPUTER VISION

**Computer Vision** is a sub-field in Artificial Intelligence that empowers machines to see, recognize and process images just the way humans do. Like other types of AI, computer vision seeks to perform and automate tasks that replicate human capabilities. In this case, computer vision seeks to replicate both the way humans see, and the way humans make sense of what they see. The main objective of computer vision is to understand the content of digital images. Computers are not the same as humans, so they don't have the gift of vision and perception.

**Machine learning** is a sub-field of artificial intelligence (AI) that focuses on developing algorithms and statistical models that enable computers to perform tasks without explicit instructions where they learn patterns and make decisions based on data thereby improving their performance overtime.

**Deep learning** is the subset of **machine learning** which is based on neural networks with many layers (deep neural networks) which are designed to automatically learn and extract features from raw data, allowing them to model complex patterns and representations.

### HOW COMPUTER VISION WORKS

Computer vision applications use input from sensing devices, artificial intelligence, machine learning, and deep learning to replicate the way the human vision system works. Computer vision applications run on algorithms that are trained on massive amounts of visual data or images in the cloud. They recognize patterns in this visual data and use those patterns to determine the content of other images.

### COMPUTER VISION APPLICATIONS

Computer vision can be combined with many types of applications and sensing devices to support a number of practical use cases. Here are just a few different types of computer vision applications:

1. **Content organization:** Computer vision can be used to identify people or objects in photos and organize them based on that identification. Photo recognition applications like this are commonly used in photo storage and social media applications.
2. **Text extraction:** Optical Character Recognition is also referred to as text recognition or text extraction which are techniques that allow you to extract printed or handwritten text from images such as posters, street signs and product labels, as well as from documents like articles, reports, forms, and invoices. The text is typically extracted as words, text lines, and paragraphs or text blocks, enabling access to digital

version of the scanned text. This eliminates or significantly reduces the need for manual data entry.

3. **Augmented reality:** Physical objects are detected and tracked in real-time with computer vision through Object Detection and Object Tracking. Object detection is used to recognize objects in visual data while Object tracking is used to understand movement, count people and objects. This information is then used to realistically place virtual objects in a physical environment.
4. **Agriculture:** Images of crops taken from satellites, drones, or planes can be analyzed to monitor harvests, detect weed emergence, or identify crop nutrient deficiency.
5. **Autonomous vehicles:** Self-driving cars use real-time object identification and tracking to gather information about what's happening around a car and route the car accordingly.
6. **Healthcare:** Photos or images captured by other medical devices can be analyzed to provide to help doctors identify problems and make diagnoses more quickly and accurately.
7. **Sports:** Object detection and tracking is used for player and strategy analysis. Computer vision technology in sports help process vast amounts of visual data received throughout the games to make match-related decisions in real-time, develop new training schemes, and much more. These computer vision solutions are of extreme help in data collection and sports analytics.
8. **Manufacturing:** Computer vision can monitor manufacturing machinery for maintenance purposes. It can also be used to monitor product quality and packaging on a production line.
9. **Spatial analysis:** The system identifies people or objects, such as cars, in a space and tracks their movement within that space. For example, you can use Azure AI Vision Spatial Analysis to detect the presence and movements of people in video. The service can do things like count the number of people entering a space or measure compliance with face mask and social distancing guidelines.
10. **Face recognition:** Computer vision can be applied to identify individuals. Facial recognition can detect and identify individual faces from an image containing one or many people's faces. It can detect facial data in both front and side face profiles.

## THE HUMAN VISUAL SYSTEM

The human visual system is a complex and sophisticated network that allows us to perceive and interpret the world around us through sight. It involves multiple structures and processes that work together to capture light, process visual information, and generate the experience of vision.

# COMPARISON OF HUMAN VISUAL SYSTEMS WITH COMPUTER VISION SYSTEMS

## 1. Sensory Input and Data Acquisition

### Human Visual System:

- **Sensory Organs:** Eyes capture light and convert it into neural signals using photoreceptors (rods and cones).
- **Dynamic Range:** The human eye has a high dynamic range, adapting to various light conditions.
- **Resolution:** Humans have a varying resolution across the visual field, with the highest resolution at the fovea (central vision).
- **Color Perception:** Humans perceive color through three types of cones (trichromatic vision) sensitive to red, green, and blue wavelengths.

### Computer Vision System:

- **Sensors:** Cameras capture images using digital sensors, converting light into electronic signals.
- **Dynamic Range:** Modern cameras have high dynamic ranges but are generally less adaptable than human eyes.
- **Resolution:** Cameras can have very high resolution, limited by the sensor's megapixel count.
- **Color Perception:** Cameras typically use RGB filters to capture color, similar to human trichromatic vision, but can also include additional filters (e.g., infrared).

## 2. Processing Architecture

### Human Visual System:

- **Parallel Processing:** Processes multiple aspects of visual information (color, motion, depth) simultaneously in different brain areas.
- **Neural Networks:** Utilizes biological neural networks in the brain, particularly the visual cortex, for processing.
- **Adaptability:** Highly adaptive to changes and capable of learning from experience and context.

### Computer Vision System:

- **Sequential and Parallel Processing:** Can use both sequential algorithms and parallel processing techniques, especially in GPU-based systems.
- **Artificial Neural Networks:** Utilizes artificial neural networks (e.g., CNNs) to process visual information, inspired by biological networks.

- **Training and Adaptability:** Requires extensive training on large datasets to adapt and improve performance.

### 3. Perception and Interpretation

#### Human Visual System:

- **Contextual Understanding:** Highly contextual, using prior knowledge, experiences, and expectations to interpret visual information.
- **Depth Perception:** Achieved through binocular vision and various monocular cues (e.g., perspective, occlusion).
- **Attention and Focus:** Capable of focusing on specific details while filtering out irrelevant information, driven by cognitive processes.

#### Computer Vision System:

- **Algorithmic Interpretation:** Relies on algorithms and models trained on labeled data to interpret images.
- **Depth Perception:** Achieved through techniques like stereo vision, depth sensors, and structure from motion.
- **Attention Mechanisms:** Implemented through attention models in neural networks, allowing the system to focus on relevant parts of the image.

## DIGITAL IMAGE FORMATION AND REPRESENTATION

**Image formation** in computer vision refers to the process of capturing and representing visual information in the form of digital images. It involves the conversion of real-world scenes into a digital format that can be processed and analyzed by computer algorithms. Understanding the image formation process is crucial for various computer vision tasks such as object detection, recognition, and tracking.

The image formation process typically involves the following stages:

1. **Scene:** The process begins with a real-world scene that contains objects, lighting conditions, and a viewing perspective. The scene can be indoor or outdoor and may involve various objects, textures, colors, and shapes.
2. **Illumination:** Illumination refers to the light sources present in the scene. It plays a vital role in determining the appearance of objects in an image. The lighting conditions, such as brightness, color, and direction, affect how the objects and their features are represented in the image.
3. **Reflectance:** Reflectance describes the interaction between light and the objects in the scene. When light interacts with an object's surface, it can be absorbed, transmitted, or reflected. The reflectance properties of objects, such as their colors, textures, and materials, influence how they appear in the image.

4. **Optics:** The next step involves the interaction of light with optical components, such as lenses, that focus the light onto a photosensitive sensor or film. The lens captures the light rays from different parts of the scene and projects them onto the sensor.
5. **Sensor:** In modern digital cameras, an image sensor is used to capture the incoming light and convert it into an electrical signal. The most common type of image sensor is a charge-coupled device (CCD) or a complementary metal-oxide-semiconductor (CMOS) sensor. These sensors consist of an array of pixels that record the intensity of light falling on them.
6. **Sampling:** The continuous analog signal captured by the image sensor needs to be converted into discrete digital values. This process is known as **Sampling**. The sensor divides the image into a grid of pixels, and each pixel measures the intensity of light falling on it. The intensity values are then quantized into discrete levels based on the bit depth of the sensor, resulting in a digital representation of the image.
7. **Quantization:** Quantization is the process of assigning discrete intensity levels to the continuous measurements obtained from the sensor. The number of intensity levels depends on the bit depth of the sensor. For example, an 8-bit sensor can represent 256 discrete levels of intensity (0 to 255).
8. **Digital Image:** The final result of the image formation process is a **digital image**, which is a two-dimensional grid of pixels, each having a specific intensity value. The image represents a visual representation of the scene captured by the camera. The resolution of the image depends on the number of pixels in the sensor and the quality of the optics.

It's important to note that image formation can also be affected by various factors such as camera settings (e.g., exposure time, aperture), noise, lens aberrations, and post-processing techniques. Understanding the image formation process helps computer vision algorithms interpret and extract meaningful information from digital images.

**Image representation** in computer vision refers to the process of converting an image into a numerical or symbolic form that can be easily understood and processed by a computer. Images are typically represented as a collection of pixels, where each pixel corresponds to a specific color or intensity value. The goal of image representation is to extract relevant features and information from the image, enabling the computer to perform various tasks, such as object recognition, image classification, and image segmentation.

There are several common techniques for image representation in computer vision:

1. **Grayscale representation:** Images are represented using a single channel where each pixel contains a grayscale value ranging from 0 (black) to 255 (white). This representation is commonly used for tasks that do not require color information.

2. **Color representation:** For color images, the most common representation is the RGB (Red, Green, Blue) format. Each pixel is represented by three color channels (R, G, and B), with each channel containing an intensity value ranging from 0 to 255.
3. **Feature extraction:** Instead of using raw pixel values, computer vision algorithms often extract relevant features from images. These features can be edges, corners, textures, or more complex representations like histograms of oriented gradients (HOG) or deep features from convolutional neural networks (CNNs).
4. **Histograms:** Image histograms represent the frequency distribution of pixel intensities in an image. They can provide valuable information about the image's contrast, brightness, and overall content.
5. **Local descriptors:** These are representations that describe local regions of an image, such as SIFT (Scale-Invariant Feature Transform) and SURF (Speeded-Up Robust Features), which are used for object recognition and matching.
6. **Global descriptors:** These representations describe the entire image and are often used for image classification tasks. Examples include bag-of-words models, histogram of gradients (HOG), and deep learning-based features like activations from pre-trained CNN models.
7. **Deep Learning-based Representations:** Convolutional neural networks (CNNs) have revolutionized image representation in recent years. CNNs automatically learn hierarchical representations from images, enabling them to capture complex patterns and features effectively.

The choice of image representation depends on the specific computer vision task at hand. Different representations may be more suitable for different applications, and the selection of the most appropriate representation significantly impacts the performance and accuracy of the computer vision system.

### **Need of Image Representation in computer vision**

Image representation plays a crucial role for the following reasons:

**Efficient storage and processing:** Raw images are typically high-dimensional data with pixel values, making them cumbersome to handle. Image representation techniques help in compressing the information into more compact and meaningful formats, reducing the memory requirements and computational complexity.

**Feature extraction:** Image representation methods extract relevant features and patterns from the images. These features highlight the important aspects of the image, such as edges, textures, shapes, and colors, which are essential for understanding and recognizing objects and scenes.

**Dimensionality reduction:** Image representation often involves dimensionality reduction techniques, which help reduce the number of features while preserving essential information. This simplifies the data and makes it easier to process, analyze, and visualize.

**Invariance and robustness:** Certain image representation techniques are designed to be invariant to specific transformations, such as rotation, scaling, or translation. This enables the computer vision algorithms to recognize objects or patterns under varying conditions, making them more robust and adaptable.

**Machine learning compatibility:** Image representations are used as inputs to machine learning algorithms for tasks like image classification and object detection. By converting images into feature vectors, they become compatible with various machine learning models, allowing us to apply a wide range of algorithms to solve specific vision problems.

**Generalization:** Effective image representations help in generalizing the learning from one set of images to others. When a model learns meaningful features, it becomes capable of recognizing similar patterns in new and unseen data, improving its performance on diverse images.

**Interpretability:** Some image representation methods aim to make the features interpretable and understandable to humans. This is particularly useful in fields like medical imaging, where visualizing and understanding the learned features are critical for decision-making and diagnosis.

## IMAGE PROCESSING

Image processing is the process of transforming an image into a digital form and performing certain operations to get some useful information from it. Image processing algorithms are used to extract information from images, restore and compress image and video data, and build new experiences in virtual and augmented reality. Computer vision uses image processing to recognize and categorize image data.

### Types of Image Processing

There are five main types of image processing:

- Visualization - Find objects that are not visible in the image
- Recognition - Distinguish or detect objects in the image
- Sharpening and restoration - Create an enhanced image from the original image
- Pattern recognition - Measure the various patterns around the objects in the image
- Retrieval - Browse and search images from a large database of digital images that are similar to the original image

### Fundamental Image Processing Steps

#### I. Image Acquisition

Image acquisition is the first step in image processing. This step is also known as preprocessing in image processing. It involves retrieving the image from a source, usually a hardware-based source.

## **II. Image Enhancement**

Image enhancement is the process of bringing out and highlighting certain features of interest in an image that has been obscured. This can involve changing the brightness, contrast, etc.

## **III. Image Restoration**

Image restoration is the process of improving the appearance of an image. However, unlike image enhancement, image restoration is done using certain mathematical or probabilistic models.

## **IV. Color Image Processing**

Color image processing includes a number of color modeling techniques in a digital domain. This step has gained prominence due to the significant use of digital images over the internet.

## **V. Wavelets and Multi-resolution Processing**

Wavelets are used to represent images in various degrees of resolution. The images are subdivided into wavelets or smaller regions for data compression and for pyramidal representation.

## **VI. Compression**

Compression is a process used to reduce the storage required to save an image or the bandwidth required to transmit it. This is done particularly when the image is for use on the Internet.

## **VII. Morphological Processing**

Morphological processing is a set of processing operations for morphing images based on their shapes.

## **VIII. Segmentation**

Segmentation is one of the most difficult steps of image processing. It involves partitioning an image into its constituent parts or objects.

## **IX. Representation and Description**



After an image is segmented into regions in the segmentation process, each region is represented and described in a form suitable for further computer processing. Representation deals with the image's characteristics and regional properties. Description deals with extracting quantitative information that helps differentiate one class of objects from the other.

## **X. Recognition**

Recognition assigns a label to an object based on its description.

### **Applications of Image Processing**

#### **Medical Image Retrieval**

Image processing has been extensively used in medical research and has enabled more efficient and accurate treatment plans. For example, it can be used for the early detection of breast cancer using a sophisticated nodule detection algorithm in breast scans. Since medical usage calls for highly trained image processors, these applications require significant implementation and evaluation before they can be accepted for use.

#### **Traffic Sensing Technologies**

In the case of traffic sensors, we use a video image processing system or VIPS. This consists of a) an image capturing system b) a telecommunication system and c) an image processing system. When capturing video, a VIPS has several detection zones which output an "on" signal whenever a vehicle enters the zone, and then output an "off" signal whenever the vehicle exits the detection zone. These detection zones can be set up for multiple lanes and can be used to sense the traffic in a particular station. Besides this, it can auto record the license plate of the vehicle, distinguish the type of vehicle, monitor the speed of the driver on the highway and lots more.

#### **Image Reconstruction**

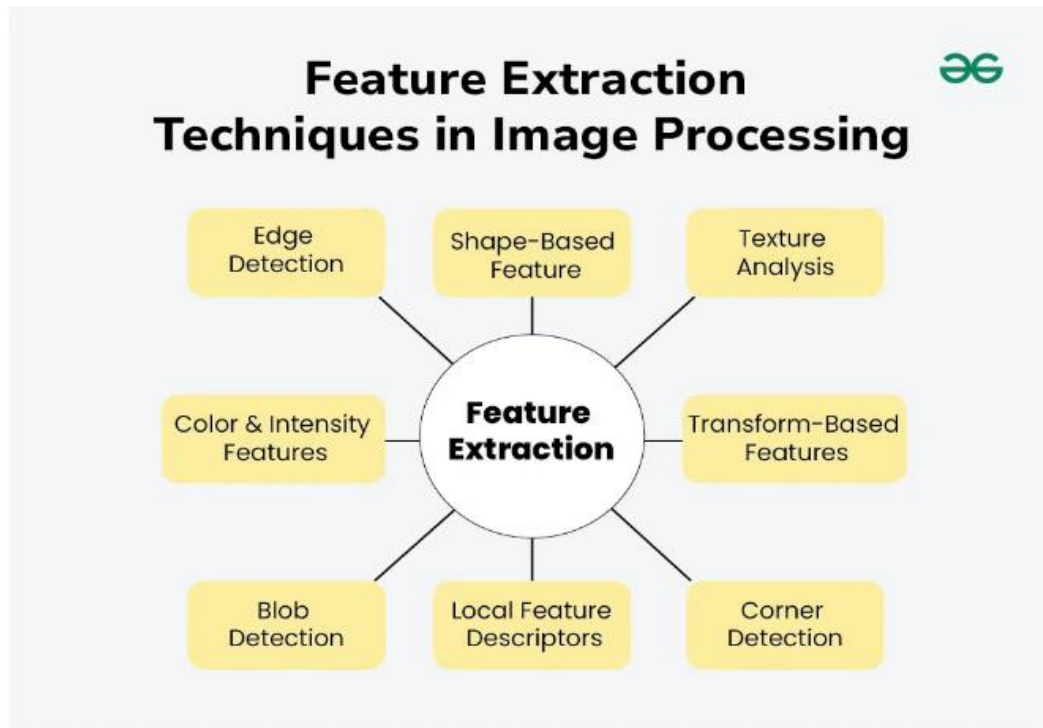
Image processing can be used to recover and fill in the missing or corrupt parts of an image. This involves using image processing systems that have been trained extensively with existing photo datasets to create newer versions of old and damaged photos.

#### **Face Detection**

One of the most common applications of image processing that we use today is face detection. It follows deep learning algorithms where the machine is first trained with the specific features of human faces, such as the shape of the face, the distance between the eyes, etc. After teaching the machine these human face features, it will start to accept all objects in an image that resemble a human face. Face detection is a vital tool used in security, biometrics and even filters available on most social media apps these days.

## Feature Extraction in Image Processing

Feature extraction is a critical step in image processing and computer vision, involving the identification and representation of distinctive structures within an image. This process transforms raw image data into numerical features that can be processed while preserving the essential information. These features are vital for various downstream tasks such as object detection, classification, and image matching.



## Feature Extraction Techniques for Image Processing

### 1. Edge Detection

Edge detection is a fundamental technique in image processing used to identify boundaries within an image. It's crucial for tasks like object detection, image segmentation, and feature extraction. Essentially, edge detection algorithms aim to locate points where the intensity of an image changes abruptly, which typically signifies the presence of an edge or boundary between different objects or regions. Common edge detection methods are:

- I. **Sobel, Prewitt, and Roberts Operators:** These methods are based on calculating the gradient of the image intensity. They operate by convolving the image with a small, predefined kernel that highlights the intensity changes in horizontal and vertical directions. By computing the gradient magnitude and direction at each pixel, these operators can identify edges where the intensity changes are significant. The Sobel operator, for example, uses a  $3 \times 3$  kernel to compute gradients, while Prewitt and Roberts operators use similar principles but with different kernel designs.

II. **Canny Edge Detector:** Unlike the previous methods, the Canny Edge Detector is a multi-stage algorithm that provides more refined edge detection. The Canny Edge Detector is known for its ability to detect a wide range of edges while suppressing noise and minimizing false detections. It comprises several steps:

- **Gaussian Smoothing:** The input image is convolved with a Gaussian kernel to reduce noise and smooth out the image.
- **Gradient Calculation:** Sobel operators are applied to compute the gradient magnitude and direction at each pixel.
- **Non-maximum Suppression:** This step helps thinning the detected edges by retaining only local maxima in the gradient magnitude along the direction of the gradient.
- **Double Thresholding:** Pixels are classified as strong, weak, or non-edge pixels based on their gradient magnitudes. A high threshold determines strong edge pixels, while a low threshold identifies weak edge pixels.
- **Edge Tracking by Hysteresis:** Weak edge pixels that are adjacent to strong edge pixels are considered as part of the edge. This helps in connecting discontinuous edge segments.

## 2. Corner detection

Corner detection is another important technique in image processing, particularly in computer vision and pattern recognition. It aims to identify points in an image where the intensity changes significantly in multiple directions, indicating the presence of corners or junctions between edges. Corners are valuable features because they often correspond to keypoints that can be used for tasks like image alignment, object tracking, and 3D reconstruction.

Common corner detection methods are:

- I. **Harris Corner Detector:** The Harris Corner Detector is a classic method for corner detection. It works by analyzing local intensity variations in different directions using the concept of the auto-correlation matrix. Specifically, it measures the variation in intensity for a small displacement of a window in all directions. By calculating the eigenvalues of the auto-correlation matrix, the algorithm identifies corners as points where the eigenvalues are large in both directions. The Harris Corner Detector typically uses a Gaussian window function to weight the intensity values within the window, which helps in making the detector more robust to noise.
- II. **Shi-Tomasi Corner Detector:** The Shi-Tomasi Corner Detector is an enhancement over the Harris Corner Detector. It uses a similar approach but introduces a different criterion for corner detection. Instead of relying solely on the eigenvalues of the auto-correlation matrix, Shi-Tomasi proposed using the minimum eigenvalue of the matrix as a corner measure. This modification leads to better performance, especially in

cases where there are multiple corners in close proximity or when the corners have varying degrees of contrast.

Both the Harris Corner Detector and the Shi-Tomasi Corner Detector are widely used in computer vision applications. They are crucial for tasks like feature-based registration, image stitching, object recognition, and motion tracking. Corner detection plays a fundamental role in extracting meaningful information from images and enabling high-level analysis and interpretation.

### 3. Blob detection

Blob detection is a technique used in image processing to identify regions within an image that exhibit significant differences in properties such as brightness, color, or texture compared to their surroundings. These regions are often referred to as “blobs,” and detecting them is useful for tasks such as object recognition, image segmentation, and feature extraction.

Common blob detection methods:

1. **Laplacian of Gaussian (LoG):** The LoG method is a popular technique for blob detection. It involves convolving the image with a Gaussian kernel to smooth it and then applying the Laplacian operator to highlight regions of rapid intensity change. The Laplacian operator computes the second derivative of the image, emphasizing areas where the intensity changes sharply. By detecting zero-crossings in the resulting Laplacian image, the LoG method identifies potential blob locations. This approach is effective at detecting blobs of various sizes but can be computationally expensive due to the convolution with the Gaussian kernel.
2. **Difference of Gaussians (DoG):** The DoG method is an approximation of the LoG method and offers a computationally efficient alternative. It involves subtracting two blurred versions of the original image, each smoothed with a Gaussian filter of different standard deviations. By subtracting the blurred images, the DoG method highlights areas of rapid intensity change, which correspond to potential blob locations. Similar to the LoG method, the DoG approach also detects blobs by identifying zero-crossings in the resulting difference image.
3. **Determinant of Hessian:** The Determinant of Hessian method is another blob detection technique that relies on the Hessian matrix, which describes the local curvature of an image. By computing the determinant of the Hessian matrix at each pixel, this method identifies regions where the intensity changes significantly in multiple directions, indicating the presence of a blob. The determinant of the Hessian measures the strength of blob-like structures, enabling robust blob detection across different scales.

These blob detection methods are valuable tools in image analysis and computer vision applications. They allow for the identification and localization of objects or regions of interest within an image, even in the presence of noise or variations in lighting conditions.

By detecting blobs, these methods facilitate subsequent processing steps, such as object tracking, segmentation, and recognition.

#### 4. Texture Analysis

Texture analysis is a vital aspect of image processing and computer vision that focuses on quantifying and characterizing the spatial arrangement of pixel intensities within an image. Understanding the texture of an image can be crucial for tasks like classification, segmentation, and recognition, particularly when dealing with images containing repetitive patterns or complex structures.

Common texture analysis methods:

- I. **Gray-Level Co-occurrence Matrix (GLCM):** GLCM is a statistical method used to capture the spatial relationships between pixels in an image. It measures the frequency of occurrence of pairs of pixel values at specified distances and orientations within an image. By analyzing these pixel pairs, GLCM can extract texture features such as contrast, correlation, energy, and homogeneity, which provide information about the texture properties of the image. GLCM is particularly effective for analyzing textures with well-defined patterns and structures.
- II. **Local Binary Patterns (LBP):** LBP is a simple yet powerful method for texture description. It operates by comparing each pixel in an image with its neighboring pixels and assigning a binary code based on whether the neighbor's intensity is greater than or less than the central pixel's intensity. These binary patterns are then used to encode the texture information of the image. LBP is robust to changes in illumination and is computationally efficient, making it suitable for real-time applications such as face recognition, texture classification, and object detection.
- III. **Gabor Filters:** Gabor filters are a set of bandpass filters that are widely used for texture analysis and feature extraction. They are designed to mimic the response of human visual system cells to different spatial frequencies and orientations. By convolving an image with a bank of Gabor filters at various scales and orientations, Gabor features can be extracted, which capture information about the texture's spatial frequency and orientation characteristics. Gabor filters are particularly effective for analyzing textures with varying scales and orientations, making them suitable for tasks such as texture segmentation, classification, and recognition.

These texture analysis methods offer valuable insights into the spatial structure and patterns present in an image, enabling more robust and informative analysis for various computer vision applications. By extracting relevant texture features, these methods facilitate tasks such as image understanding, object recognition, and scene understanding in diverse domains including medical imaging, remote sensing, and industrial inspection.

## Applications of Feature Extraction for Image Processing

- **Object Recognition:** Edges are features that are used to differentiate images from the background, textures and shapes of images are features used to differentiate between images within an image.
- **Facial Recognition:** Other factors such as facial symmetry and convexity, face shape and size, distance between eyes and distance across base of nose, size of forehead and distance across forehead, cheek and cheekbone size and paral distance, vertical height or size of face below line and between the eyes, jaw size and shape, nose size and shape, and size of the lips also have an effect on face categorisation.
- **Medical Imaging:** It is therefore evident that in medical diagnostics MRI or CT image it may possible to capture such characteristic in MRI or CT to analyze anomalies of tumors that may be caused by a disease at a high success probability.
- **Remote Sensing:** Features like Vegetation Index, water bodies and urban areas provided from the satellites are very valuable for doing the environmental mapping.
- **Content-Based Image Retrieval (CBIR):** Retrieving images from a database based on the content of the images rather than metadata.

## IMAGE SEGMENTATION

Image Segmentation is a process in computer vision and image processing that involves dividing an image into different regions or segments in order to simplify or change the representation of an image into something that is more meaningful and easier to analyze. It is used to locate objects and boundaries (lines, curves etc.) in images. To be more precise, Image Segmentation is the process of assigning a label to every pixel in an image such that pixels with the same label share certain characteristics.

Some practical applications of image segmentation are:

1. **Content-Based Image Retrieval (CBIR)** is a technique used to search and retrieve images from a database based on their visual content such as colour, texture, shape, object recognition, spatial relationships etc.
2. **Machine Vision** is the technology and methods used to provide imaging-based automatic inspection and analysis for applications such as automatic inspection, process control and robot guidance usually in the industry. Machine vision techniques enable machines to use vision in industrial tasks.
3. **Medical Imaging** is the technique and process of imaging the interior of a body for clinical analysis and medical intervention as well as visual representation of the function of some organs or tissues e.g. tumour detection, organ segmentation etc.
4. **Object Detection** is a technology that deals with detecting objects of a certain class (e.g. humans, buildings or cars) in digital images and videos. Well researched domains include face detection, pedestrian detection etc.
5. **Recognition** tasks like:

- **Face Recognition** which is a technology potentially capable of matching a human face from a digital image or a video frame against a database of faces.
- **Fingerprint Recognition** is a biometric identification method that uses unique patterns found on an individual's fingertips to verify their identity
- **Iris Recognition** is a biometric identification method that uses unique patterns in an individual's iris to verify their identity.

### TYPES OF IMAGE SEGMENTATION

Among the various segmentation types are 3 common ones which are:

1. **Semantic Segmentation:** This involves classifying each pixel in an image into specific class or category without distinguishing between different categories of the same class.
2. **Instance Segmentation:** This aims to identify and distinguish individual objects or instances within an image.
3. **Panoptic Segmentation:** is a hybrid method that combines both semantic and instance segmentation.

### IMAGE SEGMENTATION TECHNIQUES

Image segmentation techniques include:

- 1) **Thresholding:** This means separating objects based on intensity values. This method involves selecting a threshold value and classifying image pixels between foreground and background based on intensity values.
- 2) **Edge Detection:** this identifies boundaries between objects. This method identifies abrupt change in intensity or discontinuation in the image. It uses algorithms like Sobel, Canny or Laplacian edge detectors.
- 3) **Clustering Algorithm:** This method uses algorithms like K-means or Gaussian models to group object pixels in an image into clusters based on similar features like colour or texture i.e. (grouping pixels with similar characteristics).
- 4) **Watershed Segmentation:** Separating objects based on gradient magnitude i.e. The watershed segmentation treats the image like a topographical map where the watershed lines are identified based on pixel intensity and connectivity like water flowing down different valleys.
- 5) **Region-based segmentation:** This method segments the image into smaller regions and iteratively merges them based on predefined attributes in colour, intensity and texture to handle noise and irregularities in the image.

### OBJECT RECOGNITION ALGORITHM

Object recognition algorithms are computer vision algorithms that can identify and locate specific objects within an image or video by drawing a bounding box around them. Here are some various object recognition algorithms which are:

- I. **Histogram of Oriented Gradients (HOG):** This is one of the oldest methods of object detection. HOG uses a feature extractor to identify objects in an image and its function is to convert the overall size of the image into some form of array or feature vector.
- II. **SIFT (Scale-Invariant Feature Transform):** It is a computer vision algorithm used for detecting and describing local features in images. It is a broadly adopted feature extraction method in image classification tasks.
- III. **Region-based Convolutional Neural Networks (R-CNN):** This is an improvement in the object detection procedure from the previous method of **HOG** and **SIFT**. In the R-CNN models, the most essential features are extracted by making use of selective features.
- IV. **Faster R-CNN:** While the R-CNN model was able to perform the computation of object detection and achieve desirable results, there were some problems, especially the speed of the model. So, faster methods for tackling some of these issues had to be introduced to overcome the problems that existed in R-CNN.
- V. **Single Shot Detector (SSD):** It is one of the fastest ways to achieve the real-time computation of object detection tasks. SSD solves the issue by improving the frames per second to almost five times more than the Faster R-CNN model. It removes the use of the region proposal network and instead makes use of multi-scale features and default boxes.
- VI. **YOLO (You Only Look Once):** This is one of the most popular algorithms of object detection. YOLO model uses one of the best neural network archetypes to produce high accuracy and overall speed of processing. This speed and accuracy are the main reason for its popularity.

Here are several applications of object detection:

**(i) Self-Driving Vehicles**

Object detection is a fundamental technology underlying the concept of autonomous transport. Precise object recognition (pedestrians, obstacles, and other vehicles) is a prerequisite for self-driving vehicles to move safely on public roads.

**(ii) Video Surveillance**

Video surveillance systems use object detection technology to identify any abnormal activity, after which they activate automatic alerts if necessary. It is an efficient tool to provide security monitoring for financial institutions, retail establishments, and warehouses. Security services and law enforcement agencies also use similar solutions.

**(iii) Enhancing Quality Control**

Manufacturing companies use object detection technology to find defects in their products or raw materials, and by doing so, they achieve a high-automated quality assurance process. Neural networks can quickly detect even minute defects, thus completely transforming the previously entirely manual process requiring huge human resources.

**(iv) Medical Image Analysis**



According to IBM, images constitute about 90% of all medical data. Machine learning technologies capable of analyzing images to identify abnormalities and observable objects pose tremendous potential as a tool for doctors.

#### **(v) Visual Product Search**

Visual search technology allows potential buyers to scan a product in the real world and quickly find it on sale in an online store. This way, buyers feel a seamless customer experience and blur the boundaries between offline and online shopping.

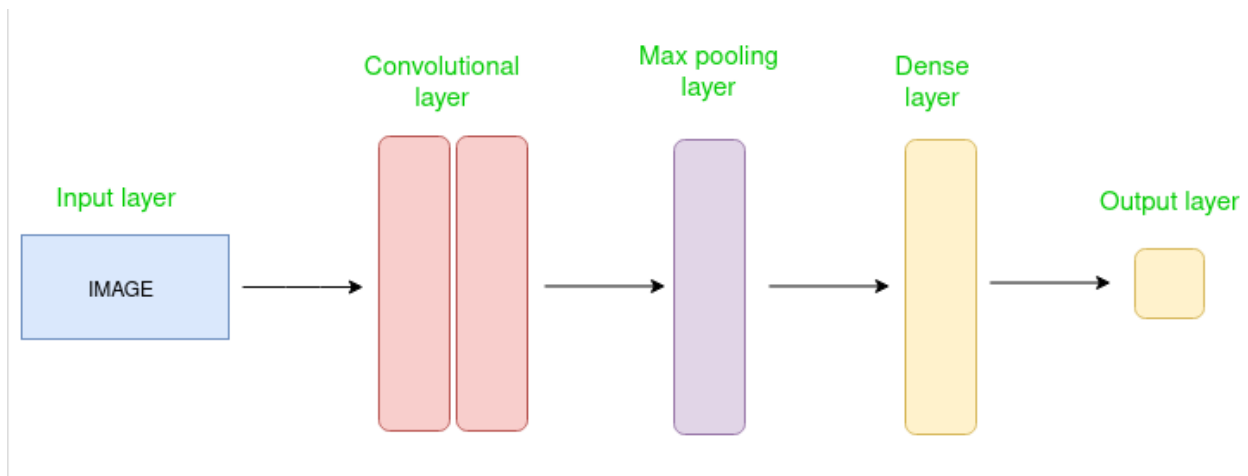
### **CONVOLUTIONAL NEURAL NETWORKS**

Convolutional Neural Networks (CNNs) are a specialized type of artificial neural network designed primarily for processing structured grid-like data, such as images. They are particularly effective in tasks like image recognition, object detection, and other computer vision-related applications. CNNs have also been adapted for use in other domains, including natural language processing and time series analysis.

In a regular Neural Network, there are three types of layers:

1. **Input Layer:** It's the layer in which we give input to our model. The number of neurons in this layer is equal to the total number of features in our data (number of pixels in the case of an image).
2. **Hidden Layer:** The input from the Input layer is then fed into the hidden layer. There can be many hidden layers depending on our model and data size. Each hidden layer can have different numbers of neurons which are generally greater than the number of features. The output from each layer is computed by matrix multiplication of the output of the previous layer with learnable weights of that layer and then by the addition of learnable biases followed by activation function which makes the network nonlinear.
3. **Output Layer:** The output from the hidden layer is then fed into a logistic function like sigmoid or softmax which converts the output of each class into the probability score of each class.

Convolutional Neural Network consists of multiple layers like the input layer, Convolutional layer, Pooling layer, and fully connected layers.



## CNN Architecture

**Convolutional layers** are the fundamental building blocks of Convolutional Neural Networks (CNNs). They apply a series of filters (or kernels) to the input data, performing a convolution operation that detects various features such as edges, textures, and patterns within the data. As these filters slide over the input, they produce feature maps that highlight the presence of these features in different parts of the input. Convolutional layers help CNNs learn hierarchical feature representations, where early layers capture basic features, and deeper layers capture more complex patterns.

**Max Pooling layer** is a common component in Convolutional Neural Networks (CNNs) used to reduce the spatial dimensions of feature maps. It works by sliding a window (typically 2x2 or 3x3) over the input feature map and selecting the maximum value within that window. This operation helps to downsample the data, reducing the number of parameters and computational load while retaining the most important features. Max Pooling contributes to making the model more robust to variations in the input, such as slight translations or distortions, by focusing on the strongest feature activations.

**Fully Connected (Dense) layers** are layers in a neural network where each neuron is connected to every neuron in the previous layer. In Convolutional Neural Networks (CNNs), after several convolutional and pooling layers have extracted features from the input, these features are flattened into a one-dimensional vector and passed through fully connected layers. These layers combine the extracted features to make final predictions. They are typically used in the final stages of a CNN, where they perform tasks such as classification or regression. The output of a dense layer is usually processed by an activation function like softmax (for classification) or linear (for regression).

Convolutional Neural Networks (CNNs) are widely used in various image analysis tasks due to their ability to automatically learn and extract features from images. Some of their applications are:

1. **Image Classification:** CNNs are highly effective for image classification due to their ability to automatically learn hierarchical feature representations from raw pixel data, making them the go-to choice for many computer vision tasks.
2. **Object detection:** Object detection using Convolutional Neural Networks (CNNs) is a process where the network not only classifies objects within an image but also localizes them by drawing bounding boxes around each detected object. This involves identifying both *what* objects are present and *where* they are located in the image.
3. **Image Segmentation:** Image segmentation is a process in which Convolutional Neural Networks (CNNs) are used to partition an image into meaningful segments or regions, typically at the pixel level. This allows the network to classify each pixel in the image as belonging to a particular class, providing a much more detailed understanding of the image compared to traditional image classification or object detection.
4. **Image Generation:** Convolutional Neural Networks (CNNs) are used in various image generation tasks, often through techniques that involve deep learning models such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and other advanced architectures.
5. **Facial Recognition:** Convolutional Neural Networks (CNNs) are widely used in facial recognition systems due to their ability to automatically learn and extract features from images.

## OpenCV

OpenCV, short for Open Source Computer Vision is a huge open-source library for computer vision, machine learning, and image processing. It plays a major role in real-time operation which is very important in today's systems. By using it, one can process images and videos to identify objects, faces, or even the handwriting of a human. It is integrated with various libraries, such as NumPy, Matplotlib, TensorFlow, PyTorch and so on in python environment. To Identify an image pattern and its various features we use vector space and perform mathematical operations on these features.

The first OpenCV version was 1.0 and hence it's free for both academic and commercial use. It has C++, C, Python, and Java interfaces and supports Windows, Linux, Mac OS, iOS and Android. When OpenCV was designed, the main focus was real-time applications for computational efficiency. All things are written in optimized C/C++ to take advantage of multi-core processing.

**Features of OpenCV** include:

- I. **Image Processing:** Filtering, thresholding, edge detection, and more.
- II. **Feature Detection:** Corner detection, blob detection, and feature extraction.
- III. **Object Recognition:** Face detection, object recognition, and tracking.
- IV. **Machine Learning:** Support for various machine learning algorithms, including SVM, decision trees, and neural networks.
- V. **Video Analysis:** Video capture, processing, and analysis.
- VI. **3D Reconstruction:** 3D modeling and reconstruction from 2D images.

OpenCV is commonly used in applications such as:

- i. Face Recognition
- ii. Object Detection
- iii. Image Segmentation
- iv. Tracking and Surveillance
- v. Autonomous Vehicles
- vi. Medical Imaging
- vii. Robotics etc.

OpenCV supports various programming languages, such as:

1. C++
2. Python
3. Java
4. Matlab

OpenCV's basic functionalities include:

- I. **Image Reading and Writing:** OpenCV can read and write various image formats, such as JPEG, PNG, TIFF, and more.
- II. **Image Processing:** OpenCV offers various image processing techniques, including:
  - Filtering (blurring, thresholding, etc.)
  - Transformations (rotation, scaling, etc.)
  - Color conversions
  - Image segmentation
- III. **Feature Detection:** OpenCV can detect and extract features from images, such as:
  - Corners
  - Edges
  - Lines
  - Circles

- Contours
- IV. **Object Detection:** OpenCV provides pre-trained models for detecting objects, including:
  - Face detection
  - Pedestrian detection
  - Car detection
- V. **Video Processing:** OpenCV can:
  - Capture video from cameras or files
  - Process video frames in real-time
  - Save video to files
- VI. **Drawing and Annotation:** OpenCV allows drawing shapes, text, and other annotations on images and videos.
- VII. **Image and Video Analysis:** OpenCV provides functions for:
  - Image histogram analysis
  - Object tracking
  - Motion analysis
  - Background subtraction
- VIII. **Machine Learning:** OpenCV includes a range of machine learning algorithms for:
  - Classification
  - Regression
  - Clustering
  - Neural networks
- IX. **Camera Calibration and 3D Reconstruction:** OpenCV offers tools for:
  - Camera calibration
  - Stereo vision
  - 3D reconstruction
- X. **Image Stitching:** Combining multiple images into a single panorama.

The modules for OpenCV include the following:

- I. **Core:** Provides basic data structures (Mat, Vec, Point, etc.) and functions for image and video processing.
- II. **ImgProc:** Offers image processing functions (filtering, thresholding, edge detection, etc.).
- III. **Video:** Enables video capture, processing, and analysis (optical flow, tracking, etc.).
- IV. **HighGUI:** Provides a simple GUI for displaying images and videos.
- V. **VideoIO:** Enables reading and writing of video files.
- VI. **ImgCodecs:** Supports image encoding and decoding.

Some of the application of OpenCV functions for image and video processing tasks are:

### **Object Detection and Recognition:**

- I. **ObjDetect:** Includes pre-trained models for object detection (face detection, pedestrian detection, etc.).
- II. **DNN (Deep Neural Network):** Supports deep learning-based object detection and recognition.

### **Feature Detection and Description:**

- I. **Features2D:** Detects and describes local features (corners, blobs, etc.) using algorithms like SIFT, SURF, and ORB.
- II. **Calib3D:** Provides functions for camera calibration, stereo vision, and 3D reconstruction.

### **Machine Learning:**

1. **ML:** Offers a range of machine learning algorithms for classification, regression, clustering, and more.

OpenCV provides numerous functions for image and video processing tasks. Some examples are:

### **Image Processing:**

#### **1. Filtering:**

- `blur()`: Blurs an image using various filters (e.g., Gaussian, median).
- `GaussianBlur()`: Applies a Gaussian filter to an image.

#### **2. Thresholding:**

- `threshold()`: Applies a threshold to an image (e.g., binary, adaptive).

#### **3. Edge Detection:**

- `Canny()`: Detects edges using the Canny algorithm.
- `Sobel()`: Applies the Sobel operator to detect edges.

#### **4. Transformations:**

- `resize()`: Resizes an image.

- rotate(): Rotates an image.
- warpAffine(): Applies an affine transformation to an image.

## 5. Color Conversions:

- cvtColor(): Converts between color spaces (e.g., RGB, HSV, grayscale).

## 6. Image Segmentation:

- watershed(): Segments an image using the watershed algorithm.

## Video Processing:

### 1. Video Capture:

- VideoCapture(): Captures video from a camera or file.

### 2. Video Writing:

- VideoWriter(): Writes video to a file.

### 3. Frame Processing:

- read(): Reads a frame from a video.
- write(): Writes a frame to a video.

### 4. Video Analysis:

- calcOpticalFlowFarneback(): Calculates optical flow between frames.
- backgroundSubtractor(): Subtracts the background from a video.

## Feature Detection and Description:

### 1. Corner Detection:

- goodFeaturesToTrack(): Detects corners in an image.

### 2. Feature Extraction:

- SIFT(): Extracts SIFT features from an image.
- SURF(): Extracts SURF features from an image.

## Object Detection:

### 1. Face Detection:

- CascadeClassifier(): Detects faces using a cascade classifier.

### 2. Object Detection:

- HOGDescriptor(): Detects objects using the HOG descriptor.

Computer vision algorithms using OpenCV can be categorized into several areas:

#### i. Image Processing:

- Filtering: OpenCV provides functions like blur(), GaussianBlur(), and medianBlur() for image filtering.
- Thresholding: threshold() function applies thresholding to separate objects from the background.
- Edge Detection: Canny() function detects edges using the Canny algorithm.

#### ii. Feature Detection and Description:

- Corner Detection: goodFeaturesToTrack() function detects corners in an image.

- Feature Extraction: SIFT (Scale-Invariant Feature Transform) **SIFT()**, SURF (Speeded-Up Robust Features) **SURF()** and ORB (Oriented FAST and Rotated BRIEF) **ORB()** functions extract features from images.

iii. **Object Detection:**

- Face Detection: `CascadeClassifier()` function detects faces using Haar cascades.
- Object Detection: `HOGDescriptor()` function detects objects using HOG features.

**Image Segmentation:**

1. Thresholding: `threshold()` function separates objects from the background.
2. Edge Detection: `Canny()` function detects edges.
3. Contour Detection: `findContours()` function detects contours.

**Tracking and Motion Analysis**

- I. Optical Flow: `calcOpticalFlowFarneback()` function calculates optical flow.
- II. Background Subtraction: `BackgroundSubtractorMOG2()` function subtracts the background.

**3D Vision**

1. Stereo Vision: `StereoBM()` function calculates disparity maps.
2. Structure from Motion: `SfM()` function reconstructs 3D scenes.

**Deep Learning**

1. Convolutional Neural Networks (CNNs): OpenCV provides a DNN module for working with CNNs.

These algorithms can be used for various applications like:

- Image and video processing
- Object recognition and tracking
- Facial recognition and analysis
- Autonomous vehicles and robotics
- Medical image analysis
- Quality inspection and defect detection.

**OpenCV Issues**

When programming with OpenCV, several common issues may arise due to differences in system configurations, image processing algorithms, or mistakes in code logic. Here are some of the issues:



Here are some common issues in OpenCV:

- 1. Installation Issues:** Problems with installing OpenCV on Windows, macOS, or Linux.
- 2. Import Errors:** Errors when importing OpenCV libraries in Python or other languages.
- 3. Image Loading Issues:** Problems loading images or videos using `imread()` or `VideoCapture()`.
- 4. Data Type Mismatch:** Errors due to mismatch between data types of images or matrices.
- 5. Null Pointer Exceptions:** Errors when trying to access or manipulate null images or matrices.
- 6. Memory Leaks:** Memory leaks due to improper deallocation of memory.
- 7. Performance Issues:** Slow performance or high CPU usage when processing large images or videos.
- 8. Incompatible Library Versions:** Issues due to incompatible versions of OpenCV or its dependencies.
- 9. Camera Capture Issues:** Problems capturing video from cameras using `VideoCapture()`.
- 10. Display Issues:** Problems displaying images or videos using `imshow()` or `namedWindow()`.
- 11. Error Handling:** Difficulty handling errors or exceptions in OpenCV code.
- 12. Version Compatibility:** Issues due to changes in OpenCV API or functionality across versions.
- 13. Thread Safety:** Issues with thread safety when using OpenCV in multi-threaded applications.
- 14. Debugging Challenges:** Difficulty debugging OpenCV code due to complex data structures or algorithms.
- 15. Documentation Limitations:** Limitations or ambiguities in OpenCV documentation.

These issues can be resolved by:

- Checking installation and import statements
- Verifying image or video file paths
- Ensuring data type compatibility
- Handling null pointers and exceptions
- Optimizing code for performance
- Ensuring library version compatibility
- Using debuggers or print statements for debugging
- Consulting OpenCV documentation and community resources

## APPLICATIONS OF MACHINE LEARNING AND DEEP LEARNING ALGORITHMS IN COMPUTER VISION

Machine learning (ML) and deep learning (DL) algorithms have numerous applications in computer vision which includes:

1. **Image classification:** Image classification is the task of assigning a label (or category) to an image based on its visual content (e.g., distinguishing between cats and dogs). It's a key problem in computer vision, often solved using machine learning techniques, particularly deep learning models such as Convolutional Neural Networks (CNNs). Applications are facial recognition, medical image diagnosis, and scene recognition.
2. **Object Detection:** Object detection is a computer vision task where an algorithm identifies and locates objects within an image or video. Unlike image classification, which assigns a single label to an image, object detection identifies multiple objects, their classes, and their locations using bounding boxes. Applications are Autonomous vehicles (detecting pedestrians and traffic signs), security surveillance, and image annotation.
3. **Image segmentation:** It is a computer vision task that involves partitioning an image into multiple segments or regions to simplify its representation and make it more meaningful for analysis. Unlike object detection, which places bounding boxes around objects, image segmentation classifies every pixel in the image to a specific class or label. Applications are Medical imaging (tumor detection), semantic segmentation (understanding scenes), and instance segmentation (differentiating between object instances).
4. **Facial Recognition:** is a biometric technology that identifies or verifies a person by analyzing facial features in an image or video. Applications include Security systems, smartphone unlock features, and social media tagging.
5. **Image Generation:** Image generation refers to the process of creating new images using algorithms and machine learning models. In recent years, deep learning models, particularly **Generative Adversarial Networks (GANs)** and **Variational Autoencoders (VAEs)**, have revolutionized this field. These models can generate realistic images from scratch, modify existing ones, or create images from textual descriptions. Applications include Art generation, style transfer, and data augmentation for training models.